

FEATURE ARTICLE

(Mis)perceptions of Other People's Social Media Use

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Do people have a clear view of *other* people's social media use? To test this, we assessed American college students' (Study 1 $N = 413$; preregistered Study 2 $N = 295$) social media frequency and perceptions of others' use across eight platforms (Facebook, Instagram, LinkedIn, Snapchat, Reddit, TikTok, Twitter/X, and YouTube). We further compared their perceptions with those in nationally representative reports. Platform frequency was positively linked to overestimating how many other people use the platform, and overestimation was positively linked to perceiving that others see the same content. These findings suggest that people make biased assumptions about who else is on social media and what they see. Our studies connect theoretical perspectives on social norms and egocentric biases to explicate (mis)perceptions of others' media behavior. In doing so, we offer insights into how social media platforms can act as windows into others' social worlds—and when those windows are more or less opaque.

Keywords: social platforms, cognitive bias, social norms, misperceived norms, social projection

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Social media platforms provide a wide range of information about others' beliefs and behaviors. People can use this information to determine the social norms in their environment, including what to believe, how to communicate, and when to respond. However, a rising body of research suggests that social media can lead people to

perceive social norms inaccurately. For example, people overestimate the amount of harmful content on social media (Lee et al., 2025) and how much people debate politics online (Bailey et al., 2024). Moreover, ideologically extreme voices dominate discourse on social media (Bail, 2022; Barberá & Rivero, 2015; Bor &

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Data Availability: Data, code, survey materials, and additional online material are available at <https://osf.io/6ztxm>. The authors' hypotheses, study design, and analytic plan were preregistered for Study 2 (<https://osf.io/fhu4m>), except Hypotheses 2 and 3b, which were not preregistered. The

authors ordered their hypotheses in this article differently from the preregistration. Hypothesis 2 in the preregistration is labeled Hypothesis 4 in this article. The data reported in this article were not used in any prior work.

Open Science Disclosures:

 The data are available at <https://osf.io/6ztxm>.

 The experimental materials are available at <https://osf.io/6ztxm>.

 The preregistered design is available at <https://osf.io/fhu4m>.

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Petersen, 2022), leading people to misperceive extreme beliefs and behavior as commonplace (Robertson et al., 2024). Decades of research have shown that social norms shape a plethora of communication processes (e.g., Chen et al., 2024; Chung & Rimal, 2016; Geber & Hefner, 2019; Postmes et al., 2000). However, the role of misperceptions in norms about how much social media platforms are used by others has been overlooked thus far.

In the present research, we unveil how social media platforms may act as windows into other individuals' social worlds—or how people assume they do. On the one hand, people may have similar behaviors and experiences on social media as one another, potentially facilitating the perception of shared social reality. On the other hand, social media ecologies have become increasingly complex (e.g., more platforms and channels; Bayer et al., 2020), potentially leading to a fragmented social reality in which few people's experiences overlap. Indeed, most people are not on most platforms (Chaffey, 2025). People adopt platforms for various motivations (Alhabash & Ma, 2017), which leads them to seek out different types of content and follow different worldviews (Lai, 2019). Coupled with algorithmic biases and self-selection effects (Brady et al., 2023), the heterogeneous social media landscape may lead users to develop an inaccurate perception of others' social media behaviors and experiences. Users may underestimate or overestimate others' use—or even project their own content exposure onto others. Our current understanding of how specific platform use creates clear or distorted windows into others' social worlds remains based on assumptions and heuristic cases (e.g., Warzel, 2023). More comprehensive approaches are needed to estimate such (mis)perceptions.

To address these research gaps, we sought to connect separate lines of research related to social norms and cognitive biases. More precisely, we integrate theoretical perspectives on the relationships between norms and social media use (Ajzen, 2020; Geber & Hefner, 2019) and the egocentric biases underlying inaccuracies in social perception (Oeberst & Imhoff, 2023; M. Ross & Sicoly, 1979). Despite the diversity of the contemporary social media ecosystem, it remains unknown whether people perceive norms differently across platforms and whether perceptions of others' use (i.e., descriptive norms) are accurate. Consequently, our understanding of representation biases—that is, the alignment between who people believe to be on social media and actual social media users—remains surface-level and subject to simple aphorisms (e.g., “social media is not real life”). Quantifying perceptions (and biases) of others' social media use is thus vital to understanding the implications of social media for social cognition and learning, including how people generalize their own views to broader social realities. We conducted exploratory (Study 1) and replication (Study 2) studies in which American college students reported their social media use frequency and perceptions of others' use and content consumption. Below, we review research on perceived norms and egocentric biases, providing theoretical background on how people perceive others' media use—and whether such perceptions are accurate.

Perceived Norms and Social Media Use

People understand what to do in a given situation based on perceived expectations of what others do, or social norms (Miller & Prentice, 1996; Sherif, 1936). Norms are a central concept across the social sciences and integral to various theoretical perspectives (Dannals & Li, 2024; Gelfand et al., 2024), including numerous

theories in media and communication research (Chen et al., 2024; Chung & Rimal, 2016; Geber & Hefner, 2019). One prominent perspective that leverages norms is the theory of planned behavior, which proposes that how people *perceive* norms explains substantial variance in their intentional behavior (Ajzen, 1991). In turn, decades of research now show that norms are a powerful driver of how people think, feel, and behave (for a review, see Gelfand et al., 2024). Social norms often outweigh other competing influences on individual behaviors. For example, ingroup norms about climate change have shown stronger links to climate behaviors than individual beliefs (Cole et al., 2022). In sum, people seek to understand social norms to communicate effectively (Chung & Rimal, 2016; Dannals & Li, 2024; Lapinski & Rimal, 2005; Rimal & Lapinski, 2015).

Communication researchers, for their part, have frequently studied how social norms influence people in the context of media use (Chung & Rimal, 2016; Fulk et al., 1990; Geber & Hefner, 2019; Wolfers, 2024; Wolfers et al., 2023). For example, studies have examined perceptions regarding when mobile phone use is appropriate in social situations (Campbell, 2007; Leuppert & Geber, 2020). Other work has looked at norms within platforms. For example, studies have examined people's perceptions of positive but inauthentic self-presentation on social media (Salopek & Eastin, 2024), whether to engage in uncivil discussion on Facebook (Mak et al., 2024), and when perceptions of appropriate behavior on a platform are more consequential than the platform's affordances (Taber et al., 2023). What is unknown is whether people have descriptively accurate perceptions of how others engage with the increasingly diverse landscape of social media platforms—such as which platforms are most commonly used—along with what underlies those (mis)perceptions.

Explicating the link between individual behavior and social norms in social media use is complicated by the potentially bidirectional nature of the relationship. In one direction, internalizing a perceived norm to use a social media platform may drive user behavior. This expectation is consistent with the theory of planned behavior (Ajzen, 1991, 2020), which suggests that a person's perceptions of media use norms predict their own use behavior. In the other direction, the more time a person spends in a context, the more likely they are to have internalized the local norms (Ryder et al., 2000). In the context of media, perceptions of others' use drive one's own (Gunther & Storey, 2003). Once a person creates a social media account, they may start to assume (or hope) others also have an account on the same platform. As individuals engage with the platform more frequently, they may also perceive it as important to more people, particularly on today's popular platforms where notifications, reactions, and algorithms all work to increase the visibility of others' use (Bayer et al., 2020). Over time, people may reciprocally use the platform even more. Both directions, therefore, suggest that there should be a positive relationship between social media norms and behaviors. Given these expectations, we hypothesized that:

Hypothesis 1: Individuals who use a social media platform more frequently will perceive that a greater percentage of people use that platform.

Egocentric Bias and (Mis)perceived Norms

A further question is whether social media frequency is linked to *inaccuracy* in perceptions about others' social media use. Prior work has shown that perceptions of social norms operate even when they

are not accurate (Gelfand et al., 2024; Prentice & Miller, 1996). Misperceived norms extend to a variety of domains in social life, including cooperation (Chatman & Flynn, 2001), prejudice (Crandall et al., 2002), public policy (Dixon et al., 2020), risky behavior (Knoll et al., 2015), and morality (Tangney et al., 2007). One contributor to misperceived norms is *egocentric bias*, in which people falsely project their own behavior and experiences onto others (Oeberst & Imhoff, 2023; M. Ross & Sicoly, 1979).

Egocentric biases may be especially likely to influence perceptions of norms on social media. Users often perceive norms as less transparent on social media than in the physical world (Dannals & Li, 2024). This lack of transparency is important because people resort to egocentric perspectives when norms are ambiguous or lack transparency (Naylor et al., 2011). Prior research suggests that people falsely assume opinions they see on their personalized feeds are representative of the broader population (Robertson et al., 2024). However, there is a lack of understanding of whether people have an accurate sense of who else is using social media—and whether such (mis)perceptions are tilted by egocentric biases.

Therefore, we sought to address three different ways that egocentric biases could inform (mis)perceptions of others' use. First, we examined whether people who use a platform more frequently are more likely to *inaccurately* perceive other people's use of that platform. To address this issue, we compared perceptions of others' social media use to nationally representative estimates to gauge the level of descriptive norm accuracy. This follows the approach taken by other studies of misperceived norms (e.g., Dixon et al., 2020; Rathje et al., 2024) and goes beyond Hypothesis 1 to determine whether social media use is linked to *bias* in perceptions of others' use. Through this approach, we sought to address whether social media frequency is related to both perceptions of others' social media and accuracy, which may be conceptually and statistically distinct. It is possible, for example, that social media frequency is related to perceptions of others' use, but not related to *inaccurate* estimates of others' use. Following the literature on egocentric biases, we hypothesized that people would generally project their own social media use onto others:

Hypothesis 2: More frequent users of platforms will overestimate the percentage of other people who use those platforms.

Second, we examined whether egocentric perceptions of others' use extends to perceptions of what others *see* on social media. Past research shows that people project not only their behavioral tendencies but also perceptual *experiences* onto others (Oeberst & Imhoff, 2023). People do this in the context of media content too, which has been referred to as presumed exposure (Gunther & Storey, 2003; Gunther et al., 2006). That is, people presume that others are exposed to the same media content that they themselves are exposed to. Whether the content people consume on social media platforms today is actually more or less similar across individuals, however, is a complex question. On the one hand, social media has increased content personalization, which may lead individuals to see content that is not representative of others' experiences (Brady et al., 2023). On the other hand, social media platforms emphasize engagement-based curation (Latzer et al., 2016), which may increase the similarity of content people are exposed to (e.g., trends), facilitating homogenization (Talaifar & Lowery, 2023). Regardless of whether actual content exposure is similar or different across users, a variety of

perspectives highlight how social media can facilitate misperceptions of others (Metzler & Garcia, 2024; Robertson et al., 2024). More generally, the popularity of social media raises potent questions about the extent to which individuals now see a shared reality (Higgins et al., 2021; Jost et al., 2018).

Compared to media use in which there is a relatively uniform experience (e.g., watching the same movie), or media use in which people select their own content independently from others (e.g., choosing articles to read in a newspaper), social media facilitates an environment that is both dynamic (e.g., algorithmically curated on a momentary basis; Metzler & Garcia, 2024) and directly linked to other users (e.g., linked to friends or embedded in communities; Carr & Hayes, 2015). Indicators of others' content exposure (e.g., quantified likes, shares) are also highly visible to users on social media, potentially more so than past media (McFarland & Ployhart, 2015). However, judgments of others' content diets based on the indicators observed within the context of users' own media use may be inaccurate (Qin, 2022). The engagement-driven nature of social media feeds today (Latzer et al., 2016) may heighten individuals' presumptions that others are seeing the "viral" content they see (Rathje & Van Bavel, 2025). Of relevance, recent work has found that presumed exposure of content on social media is egocentrically anchored (Cho et al., 2021). Accordingly, here we examine whether (mis)perceptions of others' use (i.e., what platforms other people use) carry over to perceived similarity in content exposure (i.e., what content other people see). Specifically, we hypothesized:

Hypothesis 3: People who (a) perceive and (b) overestimate that a greater percentage of others use the same social media platform that they themselves use will be more likely to believe that others see similar content as they do.

Finally, we sought to explore the relationships between egocentric perceptions of others' social media use and a previously documented egocentric bias: the false consensus effect (i.e., the tendency to project one's attributes onto others; Krueger & Clement, 1994; L. Ross et al., 1977). The idea that social media exaggerate perceptions that others share the same beliefs, values, and other characteristics has received attention (e.g., Bakshy et al., 2015; Jamieson & Cappella, 2008; Sunstein, 2009). Several studies have suggested that social media use is linked to false consensus regarding personality traits and political attitudes (Bunker & Varnum, 2021; Luzsa & Mayr, 2019; Wojcieszak, 2011). A person with high levels of false consensus may be more likely to project their social media use onto others. We therefore hypothesized that:

Hypothesis 4: The links between social media frequency and perceptions of others' social media use will be more pronounced in individuals with greater egocentric biases.

Comparing Across Social Targets and Platforms

We also sought to examine whether the hypothesized relationships varied by social targets and platforms. Perceptions of who uses social media may depend on who the perceiver is considering (e.g., someone their own age vs. the typical person). In the present research, we considered two key categories of social targets when considering others' media use: (a) peers and (b) other Americans. Social influence

is more powerful when parties share similar identities (Abrams & Hogg, 1990; Cialdini & Jacobson, 2021; Cole et al., 2022). People tend to perceive others with whom they share social identities more accurately (Chaplin & Panter, 1993; Elfenbein & Ambady, 2003) but also more egocentrically (Robbins & Krueger, 2005). Perceptions of peers' use may thus vary from perceptions of more distal targets who overlap less with the perceiver in social identity. In our studies, we asked participants to estimate their peers' and other Americans' (i.e., a more distal target) social media use.

Perceptions of others' use may also depend on the social media platform in question. Although social media platforms share common elements (e.g., digital profiles, algorithmic content feeds; Bayer et al., 2020), they vary in specific norms and affordances (Fox & McEwan, 2017; Geber & Hefner, 2019). Social media research is also plagued by the "moving target" problem, whereby researchers have many potential platforms to focus on, new platforms emerge constantly, and effects may vary widely across individuals (Gosling & Mason, 2015; Valkenburg et al., 2016). Here, we sought to examine whether the proposed effects replicated across eight established platforms in the contemporary social media ecosystem: Facebook, Instagram, LinkedIn, Snapchat, Reddit, TikTok, Twitter/X, and YouTube. We chose these platforms based on their a) popularity in the United States, especially among college-aged individuals and b) the availability of nationally representative statistics by the Pew Research Center (2024), which we could compare perceptions against.

Together, our two studies thus examined (mis)perceptions of peers' and other Americans' social media use across eight platforms. Our approach evaluated whether specific platform differences emerged in the hypothesized relationships, offering cross-platform comparisons and robustness checks. We first conducted an exploratory study during March/April 2024 (Study 1) in which participants considered four of the eight platforms (randomly assigned). We asked participants to consider four of the eight platforms in Study 1 to reduce potential survey fatigue because the study included additional exploratory measures.¹ Then, we refined and preregistered our design for Study 2 (October/November 2024), asking participants to respond to questions about all eight platforms.

Method

Participants and Procedure

We collected two samples of 608 (Study 1) and 404 (Study 2) college students at a large Midwestern university in the United States. College students are ideal starting samples, as they comprise a demographic group noted for heavy social media use and relative homogeneity in age and education level (Bolton et al., 2013; Peterson, 2001; Pew Research Center, 2024). Examining perceptions of social media use in college students is a foundation to test whether the findings replicate in other groups who may not show similar digital immersion. After filtering out participants who failed an attention check (modified from Hauser & Schwarz, 2016; see additional online material at <https://osf.io/6ztmx/>), the final samples contained 413 (Study 1) and 295 (Study 2) participants.

Demographics were similar across the two studies. In Study 1, participants ($M_{\text{age}} = 20.16$, $SD_{\text{age}} = 2.69$) were 80.3% women, 18.4% men, and 1.2% nonbinary or did not specify their gender; 70.2% White, 15.7% Asian, 8.5% Black, 6.5% Latino, 2.4% Middle Eastern, 1% American Indian, and 2.2% did not specify a listed

ethnicity. In Study 2, participants ($M_{\text{age}} = 19.95$, $SD_{\text{age}} = 2.32$) were 72.1% women, 25.9% men, and 2.0% nonbinary or did not specify their gender; 76.5% White, 13.0% Asian, 8.5% Black, 6.8% Latino, and 1.7% Middle Eastern.

Participants completed the studies online for course credit. Participants reported perceptions of others' social media use and content in a survey design with two factors. One factor was the target: Participants estimated how many of their (a) peers (i.e., "people your own age") and (b) other Americans used social media and what content they saw there. The second factor was the platform: Participants estimated others' social media use and content for the following platforms: Facebook, Instagram, LinkedIn, Snapchat, Reddit, TikTok, Twitter/X, and YouTube. In Study 1, participants made estimations for only four platforms to minimize survey fatigue in a longer survey with exploratory measures. However, in Study 2, we refined our design from Study 1 by asking participants to report perceptions of others' social media use and content similarity on all eight platforms (rather than just four). Presentation orders of the factor levels were randomized across participants.

Measures

Social Media Frequency

We used a frequency measure to assess social media use. Specifically, participants reported how often they used the social media platforms on an 8-point scale (1 = *never* to 8 = *hourly*). Frequency measures are a common way to assess communication technology use (for an overview, see Boase & Ling, 2013; Kuru et al., 2017), and there are reasons why a frequency measure aligns well with the present research relative to other metrics (e.g., how much time people spend on social media). First, the time people spend on a platform often comprises automatic behaviors that are performed with a low level of awareness (Bayer & Campbell, 2012; Bayer et al., 2022). Thus, users are likely to be more aware of their social media frequency (e.g., "did I check Instagram today?") compared to duration (e.g., "how much time did I spend on Instagram today exactly?"). Indeed, frequency measures have been shown to fare better than duration reports in correspondence to logged use (Boase & Ling, 2013). In addition, frequency more clearly aligns with measures in nationally representative survey data like the Pew Research Center (2024) estimates we sought to compare participants' perceptions of others against.

Perceptions of Others' Use

To assess perceptions of others' social media use, we asked participants to estimate what percentage of their peers and other Americans use the social media platforms (defined as "at least once a month") on a 0%–100% scale. They reported these perceptions regarding the same platforms on which they reported their own frequency.

¹ In Study 1, for exploratory purposes, participants also estimated the impact of social media (e.g., on political polarization and mental health). They also reported perceptions of their friends', democrats', and republicans' social media use. See additional online material at <https://osf.io/6ztmx/> for a full list of the measures.

Perceptions of Others' Content

To assess perceptions of others' content, we instructed participants to estimate the percentage of their peers and other Americans who see similar content on social media from 0 (*none of this group sees similar content to me*) to 100% (*all of this group sees similar content to me*). They reported these perceptions regarding the same platforms they were assigned to report their frequency and perceptions of others' use (i.e., four platforms in Study 1 and eight platforms in Study 2). If a participant did not use a platform they were assigned to (i.e., answered "never" on the frequency measure), we omitted their perceptions of others' content from our analyses. Exploratory information about the types of content participants consumed is available in the additional online material at <https://osf.io/6ztmx/>.

Egocentric Biases

To assess more general egocentric biases, we used the truly false consensus effect procedure (Krueger & Clement, 1994), which allows researchers to examine false consensus effects in established scales. We assessed false consensus effects in terms of two domains: personality traits and political attitudes. We used 10 items from the Big Five Inventory-2 (e.g., "I am a person who tends to be quiet"; Soto & John, 2017) for personality traits, and 10 items covering different political attitudes (e.g., attitudes toward legalized euthanasia, protection of the environment over economic growth, social welfare programs; Bauman & Geher, 2002; Luzsa & Mayr, 2019). Participants indicated binary endorsement with items (i.e., either endorsing the item or not), and then estimated what percent of others (out of 100%) would endorse the same items. We specified the "others" as other college students so that participant estimations could be compared to actual endorsement rates reported in our sample. Then, we subtracted the average endorsement in our sample from each participant's estimate. Each participant received a difference score for each item representing their over- or underestimation of other participants' endorsements. Finally, we calculated truly false consensus effects by obtaining the within-person correlation between each participant's endorsement of the items and the difference scores representing their level of over or underestimation. Each participant received two truly false consensus effect scores, one for personality traits and one for political attitudes. A higher score indicates that the participant demonstrated greater false consensus in the relevant domain.²

Analytic Approach

To aid readers, we use italics below to distinguish *raw perceptions* (relevant to Hypotheses 1 and 3a) from *accuracy* of the perceptions (relevant to Hypotheses 2 and 3b). *Raw perceptions* refer to the estimate of the percentage of others who use each social media platform. *Accuracy* of the perceptions refers to those estimates after subtracting the percentage of others who actually use each platform as reported by the Pew Research Center during the same year as our data collection (i.e., Pew Research Center, 2024; see additional online material at <https://osf.io/6ztmx/> for the Pew report statistics). In cases where participants' perceptions were inaccurate, we use the terms "overestimation" or "underestimation" to denote the direction (i.e., whether participants perceived a higher

vs. lower percentage of others who use a platform relative to the Pew report).

Our survey data were nested, which violates the assumption of independent observations. To provide a more robust test of the hypotheses, we ran 24 multilevel linear mixed-effects models for each study, each specifying platform ($k = 8$) nested within participants (Study 1, $n = 413$; Study 2, $n = 295$).

We ran eight models to test Hypotheses 1, 2, 3a, and 3b. Specifically, we tested two separate models for each of these four hypotheses to show effects regarding perceptions of peers separately from perceptions of other Americans. We first ran the linear mixed-effects models to examine how frequency related to *raw perceptions* of others' use (Hypothesis 1) and *accuracy* of those perceptions (Hypothesis 2), with random intercepts for participants and random intercepts and slopes for the platforms. Next, we ran the linear mixed-effects models to examine how (Hypothesis 3a) *raw perceptions* of others' use and (Hypothesis 3b) *accuracy* of those perceptions related to perceptions of content similarity with random intercepts for participants and random intercepts and slopes for the platforms.

We then conducted the remaining 16 models to test Hypothesis 4. Specifically, we re-ran all eight of the previous linear mixed-effects models to test whether each of the two false consensus domains (personality and political attitudes) moderated the hypothesized relationships (Hypothesis 4). That is, we ran models that paralleled those above (Hypotheses 1, 2, and 3), with an added interaction term representing the moderating effect of each of the two false consensus domains.

Following recommendations to address Type I error rates (Barr et al., 2013; Heisig & Shaeffer, 2019), we employed a maximal random effects structure in our models. All continuous predictor variables were centered around the grand mean. We used the lmer function with restricted maximum likelihood estimation in the lme4 package in RStudio 4.4.0 to test the multilevel linear mixed-effects models. We conducted these analyses for each of our two studies. Generally, results were consistent between the two studies. We note the few cases when key findings were different between studies.

Results

Descriptive Statistics

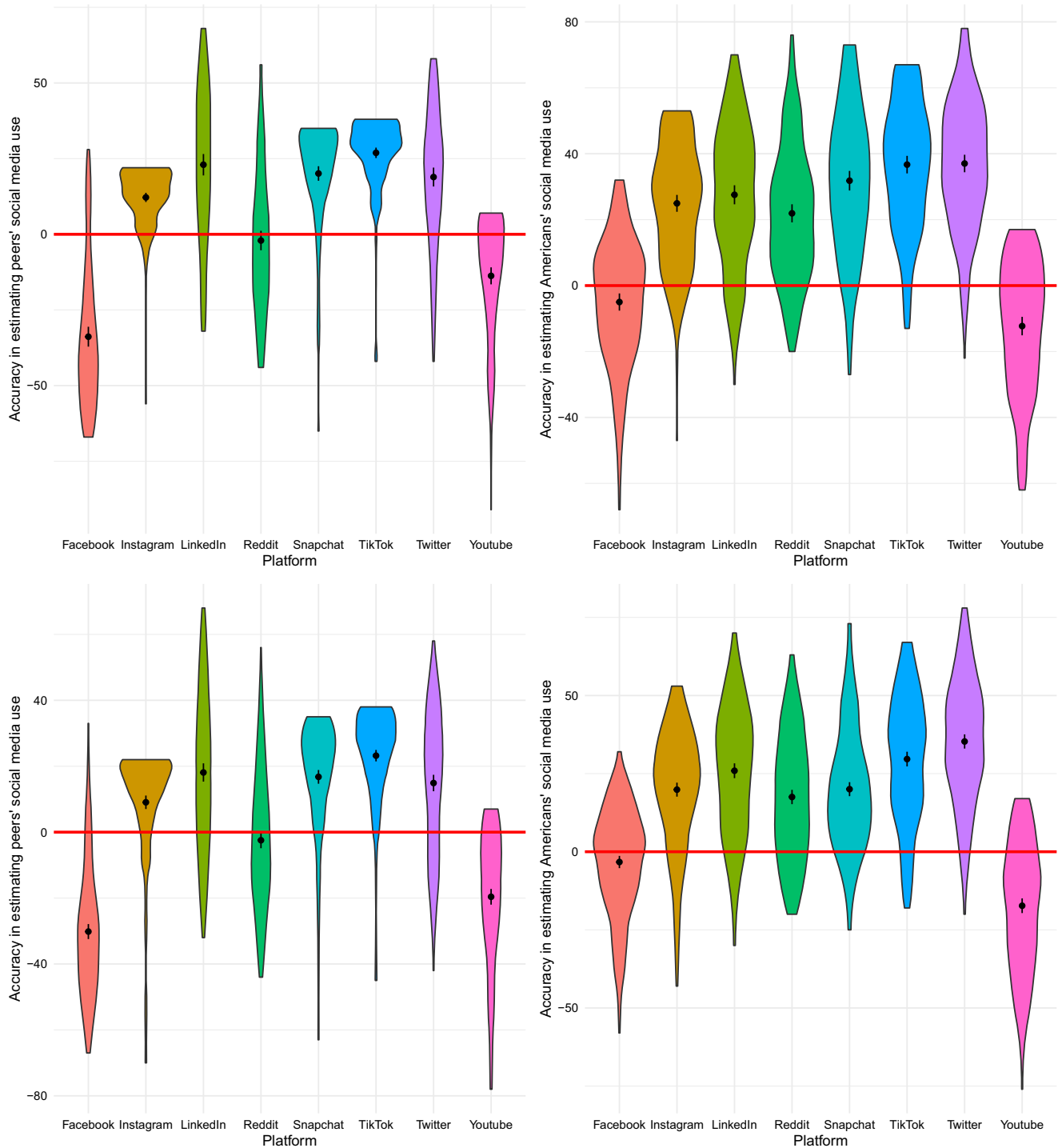
Participants underestimated the percentage of their peers and Americans who used Facebook and YouTube (and the percentage of their peers who used Reddit) but overestimated the percentages of peers and Americans who used the other platforms (see Figure 1). We confirmed the statistical significance of these differences (see additional online material at <https://osf.io/6ztmx/>). Full tables of the descriptive statistics and zero-order correlations between study variables are shown in the additional online material at <https://osf.io/6ztmx/>.

Before testing our hypotheses, we first examined whether multilevel modeling would be appropriate. To do so, we calculated intraclass correlation coefficients (ICCs), which suggest whether variance in the intended outcome variables in our models was

² For exploratory purposes, we also included a measure of bias blindspots (Scopelliti et al., 2015), which is another type of egocentric bias. However, the reliability of scores from this measure was low (see additional online material at <https://osf.io/6ztmx/>). Therefore, we did not report analyses regarding bias blindspots.

Figure 1

Accuracy in Estimating Peers' (Left Panels) and Americans' (Right Panels) Social Media Use by Platform in Study 1 (Top Panels) and Study 2 (Bottom Panels)



Note. Red line indicates 100% accuracy. A positive value indicates overestimations. A negative value indicates underestimation. Line within violins denotes 95% CI around the mean. Raw estimates and accuracy descriptives are reported in the additional online materials at <https://osf.io/6ztmx/>. CI = confidence interval.

attributable to the two nesting factors: platform and individual participant. We calculated the ICCs by specifying null models estimating the amount of variance in *raw perceptions* and *accuracy* of the perceptions of others' social media use attributable to platforms and individual participants. Even ICCs as low as .050 can warrant the use of multilevel modeling to account for variance attributable to nesting factors (Huang, 2018). The ICCs in the null models (see Table 1) suggested that a substantial amount of variance in *raw perceptions* of peers' and Americans' use, *accuracy* in estimating peers' and Americans' use, and perceptions of content seen by peers and Americans was attributable to the nesting factors. These findings suggest that multilevel modeling was appropriate to test our hypotheses. In addition, when decomposing the ICCs by each nesting factor, the results suggested that some variance in the intended outcome variables was attributable to platform (not just individual participant); thus, we included platform as a nesting factor in our models in addition to nesting within individual participants.

Linear Mixed-Effects Models

We observed significant positive fixed effects of frequency on *raw perceptions* of others' social media use for both peer and American models in both Study 1 and Study 2 (Table 2). Specifically, frequency was positively related to *raw perceptions* of others' social media use, but the size of this relationship varied meaningfully across platforms and targets (see Figures 2 and 3; top panels). The fixed effects suggested that the relationships were stronger regarding perceptions of peers' use compared to other Americans' use. Random slopes suggested that in Study 2, Twitter/X was the only platform to show a negative relationship between frequency and *raw perceptions* of other Americans' social media use. All slopes were positive across platforms in Study 1. Taken together, these results supported Hypothesis 1 but suggested that relationship strength varies across platforms and targets.

We observed significant fixed effects of frequency on *accuracy* in estimating others' social media use for both peer and American models in both Study 1 and Study 2 (Table 3). The fixed effects suggested that the relationships were stronger regarding perceptions of peers' use compared to other Americans' use. Random slopes suggested that frequency was positively related to overestimation of others' social media use for most but not all platforms (see Figures 2 and 3; bottom panels). In particular, two exceptions showed that more frequent YouTube and Facebook users were actually more *accurate* at estimating others' use of these platforms in both studies. Another exception occurred in Study 2 only, whereby Twitter/X was the only platform to show a negative relationship between frequency and

overestimations of other Americans' social media use. Taken together, these results supported Hypothesis 2 across targets and occurred for most (but not all) of the eight included platforms.

We observed significant fixed effects of *raw perceptions* of others' social media use on perceptions of *content* seen by others for both peer and American models (Table 4). That is, individuals who estimated that a higher percentage of others use a platform were more likely to believe others saw similar content to them on that platform as well. This effect was relatively consistent across platforms (see additional online material at <https://osf.io/6ztmx/>). These results supported Hypothesis 3a and suggested that the effects were mostly robust across platforms and targets.

We observed significant fixed effects of *accuracy* in estimating others' social media use on perceptions of *content* seen by others in both peer and American models (Table 5). That is, individuals who *overestimated* others' use on a platform were more likely to believe others saw the same content on that platform as well. Again, the fixed effects also suggested that the relationships here were stronger regarding perceptions of peers' use compared to other Americans' use. Again, the effect was relatively consistent across platforms (see additional online material at <https://osf.io/6ztmx/>). These results supported Hypothesis 3b and suggested that the effects were mostly robust across platforms and targets.

Our tests of whether egocentric bias moderated the hypothesized relationships showed some deviations between studies. In Study 1, the false consensus measures did not significantly moderate most of the previously hypothesized relationships (p 's > .05). There was one exception whereby greater false consensus in political attitudes attenuated the links between frequency and *raw perceptions* of other Americans' use ($\beta = -.05$, $p = .016$). In Study 2, most of the interaction effects were insignificant as well, but two significant moderation effects appeared: Greater false consensus in personality traits attenuated the links between a) frequency and *raw perceptions* of other Americans' use ($\beta = -.05$, $p = .006$) and between b) frequency and *accuracy* of the perceptions of other Americans' use ($\beta = -.05$, $p < .001$). Taken together, these findings failed to support Hypothesis 4. Further, patterns in the significant interactions suggested the opposite of what was predicted in Hypothesis 4 (see additional online material at <https://osf.io/6ztmx/> for visualizations).

Discussion

We conducted two studies on American college students examining the links between social media frequency and perceptions of other people's use across eight platforms. We also assessed

Table 1
Intraclass Correlation Coefficients for Outcome Variables by Nesting Factor and Study in Null Models

Outcome variable	Overall ICC		Platform ICC		Individual ICC	
	Study 1	Study 2	Study 1	Study 2	Study 1	Study 2
Peers' use (raw perception)	.628	.639	.535	.499	.093	.140
Americans' use (raw perception)	.494	.507	.184	.197	.310	.310
Peers' use (accuracy)	.621	.636	.525	.488	.096	.148
Americans' use (accuracy)	.672	.658	.470	.442	.202	.215
Content seen by peers	.447	.487	.104	.132	.343	.355
Content seen by Americans	.529	.475	.033	.053	.495	.422

Note. Overall ICC = variance attributable to both nesting factors; Platform ICC = variance attributable to just platform; Individual ICC = variance attributable to just individual participant; ICC = intraclass correlation coefficient.

Table 2*Predicting Raw Perceptions of Others' Social Media Use With Frequency*

Model estimate												
Effect	Peer						American					
	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>
Fixed effects												
Study 1												
Intercept	67.68	6.01		7.05	11.26	<.001	61.57	2.92		7.39	21.12	<.001
Frequency	8.52	1.07	.29	6.67	7.94	<.001	3.46	0.80	.16	5.71	4.31	.006
Study 2												
Intercept	63.87	5.35		7.18	11.93	<.001	57.38	3.27		7.67	17.53	<.001
Frequency	8.11	1.24	.30	7.34	6.56	<.001	3.18	0.74	.15	6.69	4.30	.004
Effect	Variance				<i>SD</i>		Variance				<i>SD</i>	
Random effects												
Study 1												
Intercept (individual)	88.68				9.42		151.82				12.32	
Intercept (platform)	285.31				16.89		63.46				7.97	
Slope (platform)	6.47				2.54		2.73				1.65	
Residual	288.70				16.99		239.94				15.49	
Study 2												
Intercept (individual)	114.90				10.72		151.63				12.31	
Intercept (platform)	224.84				14.99		80.41				8.97	
Slope (platform)	10.41				3.23		2.69				1.64	
Residual	251.89				15.87		233.16				15.27	

Note. *N* = 412 (Study 1); *N* = 295 (Study 2). *SE* = standard error.

whether people had accurate perceptions of others' social media use by comparing those perceptions to nationally representative estimates. Overall, we found general support for three out of our four hypotheses, offering new insights into the roles of descriptive norms and egocentric biases in conjunction with social media use. First, we found that more frequent social media users were more likely to perceive—or perhaps presume—that other people use the same platforms as themselves. This finding supported Hypothesis 1 that individuals who use a platform more frequently are more likely to believe that a greater percentage of other people will use that platform. The findings are consistent with the theory of planned behavior (Ajzen, 1991, 2020) and the notion that the more time a person spends in a social context, the more they internalize local norms (Ryder et al., 2000). Second, we found that social media frequency was positively linked to *overestimating* others' social media use. This finding supported Hypothesis 2 and reaffirms that people appear to project their own use onto others in ways that are often inaccurate. In turn, our results are indicative of egocentric biases (Oeberst & Imhoff, 2023; M. Ross & Sicoly, 1979). Third, we found that perceptions and overestimates of others' social media use were positively correlated with egocentric perceptions that other users see similar content, supporting Hypothesis 3. These findings corroborate prior research suggesting that users map their own social media experiences onto others (e.g., Cho et al., 2021) and show, by extension, that they do so in a biased way. Fourth, we did not observe the anticipated moderating effects of more general egocentric biases on the link between social media use and perceptions of others' use, in contrast to Hypothesis 4.

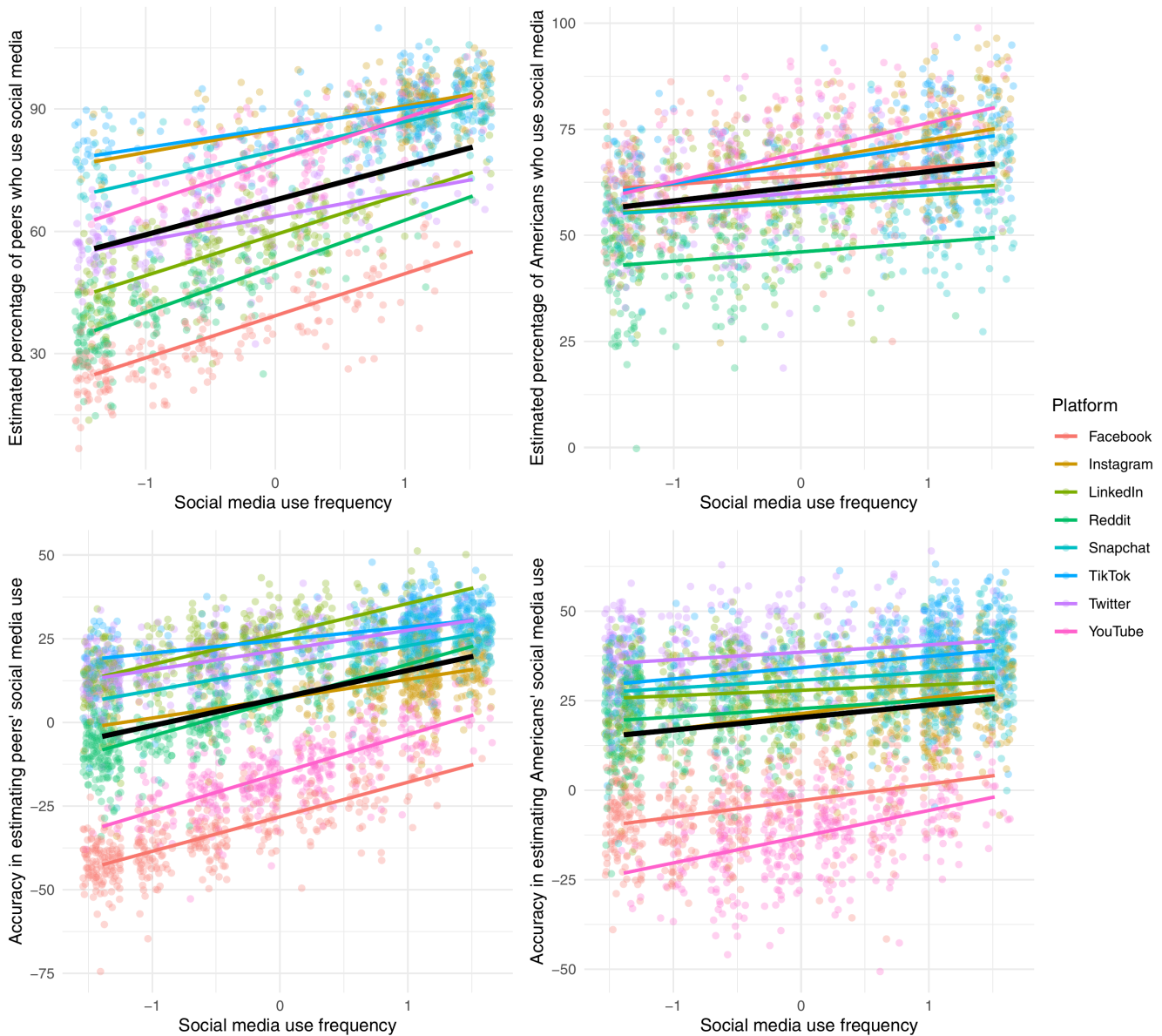
The findings were relatively robust across platforms. This is noteworthy given the wide range of documented platform features, affordances, and psychological effects in social media research (see Bayer et al., 2020; Fox & McEwan, 2017). This is also notable since a

paucity of work on norms in media and communication research considers the role of platform or channel differences, which is increasingly vital in a scattered media landscape. A few exceptions to the cross-platform consistency in the findings should be highlighted, however. First, more frequent Twitter/X users believed other Americans were less likely to use the platform as themselves in Study 2. Of note, the present studies occurred amid major changes to the platform (i.e., ownership, rebranding as X), which coincided with shifting public perceptions of the platform (Hatch, 2024), potentially impacting the present results. We also observed that, for some platforms (i.e., Facebook and YouTube), frequency was positively linked to more accurate perceptions of others' social media use. Interestingly, participants tended to underestimate others' use of these platforms. Hence, we provide initial evidence that more frequent use may result in more accurate perceptions of platforms such as Facebook and YouTube that are commonly used. Such cases suggest how egocentrism may sometimes lead to more accurate views of norms when inhabiting popular social spaces. Future research may also examine whether some platform-specific features or affordances tilt egocentric biases.

The findings were also relatively similar across social targets (peers vs. Americans). However, the effect sizes were more pronounced in perceptions of peers. Specifically, we observed stronger relationships for peers (vs. other Americans) between social media frequency and both (a) perceptions of others' use and (b) the similarity of content they see. This is consistent with prior research suggesting that although people are more accurate at perceiving similar others (Chaplin & Panter, 1993), they are also more likely to egocentrically project their own attributes onto them (Robbins & Krueger, 2005). Social projection theory suggests that egocentric bias enhances cooperation within ingroups (Krueger, 2013). An implication in the case of social media is that more frequent users

Figure 2

Predicting Raw Perceptions of Others' Social Media Use (Top) and Accuracy of Those Perceptions (Bottom) by Platform (Study 1)



Note. Frequency on the x-axis reflects scaled values. Black line indicates fixed effect.

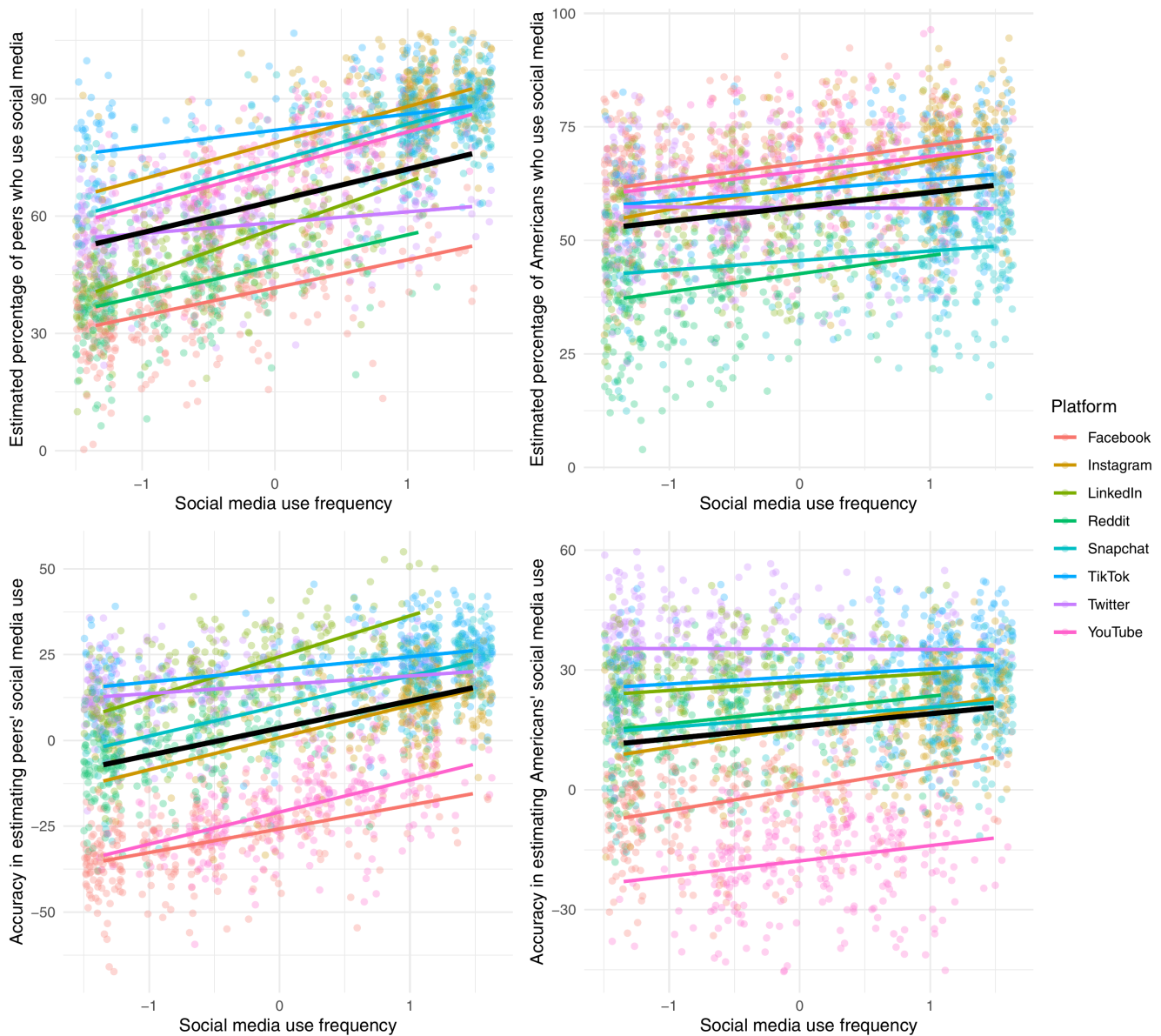
may project their own use and content exposure more so onto similar others than outgroups. Egocentric projection of social media behaviors and experiences toward ingroups may potentially exacerbate outgroup animosity observed on social media (e.g., Rathje et al., 2021). Indeed, users are more likely to egocentrically project what they see on social media if they identify with the content and perceive it as reflecting reality (Cho et al., 2021). Future research can thus investigate how self-other overlap in egocentric social judgment extends more broadly to (mis)perceptions derived from social media use.

Other findings raise the question of whether misperceptions about others' social media use reflect a general egocentric bias or a distinct egocentric bias not found in other perceptual domains. This was

complicated by the lack of significant effects related to the moderating role of general egocentric biases (false consensus measures). In Study 2, we even observed a few interactions whereby participants with greater egocentric bias showed *weaker* projection of personal use onto other Americans (i.e., the opposite of Hypothesis 4). In other words, people who scored higher on false consensus in terms of personality traits showed the same level of perceptions of other Americans' use regardless of how frequently they use social media. Such individuals may project their adoption of a platform onto others even if they seldom check their social media. Notably, we also failed to find a strong direct relationship between egocentric biases and misperceptions of others' social media use (see additional online material at <https://osf.io/6ztmx/>), undermining the idea that the more

Figure 3

Predicting Raw Perceptions of Others' Social Media Use (Top) and Accuracy of Those Perceptions (Bottom) by Platform (Study 2)



Note. Frequency on the x -axis reflects scaled values. Black line indicates fixed effect.

general biases directly increase (vs. moderate) misperception. On the other hand, people who are especially high on egocentric biases may not share normative misperceptions that others have. However, given the inconsistency in these effects, future research is needed to unravel how egocentric biases regarding social media use connect to individual differences in egocentric biases in other domains.

Theoretical Implications

More broadly, our studies illustrate how social media platforms may act as windows into other people's social worlds—or how people assume they do—and when those windows are more or less

opaque. In our studies, we focused primarily on how the frequency of using a given platform may anchor perceptions of others' use of that platform. The more a person frequents a platform, the more they look through that platform-specific window to understand the world. In this way, platform frequency may act as a lens that shapes one's understanding of who else is looking through it too (i.e., using that platform to see the social world). Our studies provide repeated evidence that people tend to think others are looking through the same windows as they are (with a few exceptions, as discussed above). Moreover, beyond influencing perceptions of others' general use, we explored how personal use shapes what people think others *see* in the window—that is, whether they see similar or different

Table 3*Predicting Overestimations of Others' Social Media Use With Frequency*

Effect	Model estimate											
	Peer						American					
	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>
Fixed effects												
Study 1												
Intercept	7.28	6.98		7.06	1.04	.331	20.33	6.54		7.12	3.11	.017
Frequency	8.25	1.14	.29	6.83	7.24	<.001	3.48	0.82	.13	6.00	4.25	.005
Study 2												
Intercept	3.61	6.63		7.14	.54	.603	15.89	6.23		7.18	2.55	.037
Frequency	7.92	1.26	.29	7.31	6.29	<.001	3.14	0.71	.12	6.47	4.45	.004
Effect	Variance				<i>SD</i>		Variance				<i>SD</i>	
Random effects												
Study 1												
Intercept (individual)			86.90			9.32			152.69			12.36
Intercept (platform)			385.55			19.64			337.52			18.37
Slope (platform)			7.54			2.75			2.98			1.73
Residual			284.02			16.85			239.59			15.48
Study 2												
Intercept (individual)			116.84			10.81			151.41			12.31
Intercept (platform)			347.29			18.64			305.50			17.48
Slope (platform)			10.85			3.29			2.36			1.54
Residual			244.59			15.64			233.32			15.28

Note. *N* = 407–412 (Study 1); *N* = 291–295 (Study 2). *SE* = standard error.

content. In sum, our data provide initial evidence that more frequent users project their use onto others, and in turn, egocentrically believe others see the same things as they do. On the one hand, these windows in social realities could clarify users' views of others'

behaviors and experiences. Alternatively, these windows could distort their perceptions of what others do and see online.

Nonetheless, future research is needed to clarify the conceptualization and operationalization of social media (mis)perceptions.

Table 4*Predicting Perceptions of Content Similarity With Raw Perceptions of Others' Social Media Use*

Effect	Model estimate											
	Peer						American					
	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>
Fixed effects												
Study 1												
Intercept	56.63	1.56		8.13	36.41	<.001	41.79	1.29		12.35	32.51	<.001
Others' use estimate	7.28	0.91	.31	5.00	8.01	<.001	3.94	.56	.19	937.56	7.04	<.001
Study 2												
Intercept	56.73	1.85		10.25	30.71	<.001	38.75	1.87		10.00	20.73	<.001
Others' use estimate	8.18	0.76	.33	10.37	10.77	<.001	3.47	1.00	.16	8.61	3.48	.007
Effect	Variance			<i>SD</i>			Variance			<i>SD</i>		
Random effects												
Study 1												
Intercept (individual)			168.26			12.97			186.72			13.66
Intercept (platform)			13.28			3.64			8.17			2.86
Slope (platform)			2.41			1.55			0			.03
Residual			305.77			17.49			213.33			14.61
Study 2												
Intercept (individual)			181.26			13.47			186.57			13.66
Intercept (platform)			20.36			4.51			21.35			4.62
Slope (platform)			1.36			1.16			5.63			2.37
Residual			310.35			17.62			263.51			16.23

Note. *N* = 412 (Study 1); *N* = 295 (Study 2). *SE* = standard error.

Table 5*Predicting Perceptions of Content Similarity With Overestimations of Others' Social Media Use*

Model estimate												
Effect	Peer						American					
	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>	<i>b</i>	<i>SE</i>	β	<i>df</i>	<i>t</i>	<i>p</i>
Fixed effects												
Study 1												
Intercept	55.74	2.59		7.75	21.50	<.001	41.47	1.50		10.17	27.56	<.001
Others' use overestimate	7.28	0.83	.29	662.20	8.74	<.001	4.68	0.72	.22	30.50	6.46	<.001
Study 2												
Intercept	54.79	2.78		8.16	19.69	<.001	37.95	1.83		10.88	20.70	<.001
Others' use overestimate	8.64	0.81	.33	10.91	10.72	<.001	4.09	1.22	.18	8.42	3.36	.009
Effect	Variance			<i>SD</i>			Variance			<i>SD</i>		
Random effects												
Study 1												
Intercept (individual)		169.73			13.03		188.10			13.71		
Intercept (platform)		48.37			6.95		13.01			3.61		
Slope (platform)		0			.01		.57			.75		
Residual		306.27			17.50		212.59			14.58		
Study 2												
Intercept (individual)		185.04			13.60		187.27			13.69		
Intercept (platform)		55.21			7.43		19.83			4.45		
Slope (platform)		1.15			1.07		8.58			2.93		
Residual		316.37			17.79		263.35			16.23		

Note. *N* = 406–412 (Study 1); *N* = 291–295 (Study 2). *SE* = standard error.

One way of understanding misperceived norms is a form of representation perception (or bias). That is, social environments—whether offline or online—provide lenses that shape our representation of broader social worlds. In the present study, we focused on how social platform use may relate to people's cognitive representations of who is on a particular platform. However, there are a variety of other social realities that people are likely to infer from social media—including certain communities (e.g., subreddits), identity groups (e.g., gamers), and geographic populations (e.g., Americans). For example, whether a person falsely perceives extreme ideological posts as representative of what most people believe depends on whether they mistake the posters as representative of the general population. Hence, future work is needed to explicate the cognitive representations of different social entities that emerge from social media, particularly given the variability in social landscapes that people roam online. In doing so, researchers should attend to not only what people think others use or believe, but also what others see. After all, visibility is its own form of power in a social (media) world. In extrapolating from what is seen on social media, a person makes assumptions about who (or who did not) see it with them. By carefully considering the mechanisms through which people observe a social environment (e.g., TikTok) and generalize it to a broader social reality (e.g., Canadians), researchers can help to explicate the cognitive assumptions in research on misperceptions derived from social media use.

While our studies help to link theoretical perspectives on social norms and egocentric biases tied to social media, follow-up studies may benefit by also accounting for content and cultivation perspectives. People use social media for various reasons (Alhabash & Ma, 2017) and consume a wide variety of content types (Hallinan et al., 2023). The diversity of user content diets highlights the fragmentary nature of the contemporary media landscape, particularly when compared to earlier communication systems with more uniform

environments and experiences (Bayer et al., 2020; Carter et al., 2023). Cultivation effects whereby users project the types of content they see on media onto their broader realities (Gerbner & Gross, 1976) have been observed on social media (Hermann et al., 2023). Although cultivation perspectives have been critiqued (see Morgan & Shanahan, 1997), future research may consider how (mis)perceptions of others' use are relevant to cultivation effects. For example, cultivation effects are stronger when users reason heuristically (Ellithorpe et al., 2016). Social media often display ambiguous norms (Dannals & Li, 2024), inviting heuristic reasoning in which people default to their own experience (Naylor et al., 2011). Accordingly, user-specific content diets on social media may cultivate especially specific (and egocentric) conceptions of social reality.

Limitations and Future Directions

Several limitations of our research should be noted. First, our cross-sectional, one-timepoint studies cannot draw causal inferences. Open questions remain regarding the causal relationship between social media frequency and (mis)perceptions of others' social media behaviors and experiences. Future research may conduct experiments or longitudinal studies to untangle the directionality and temporal stability of media use perceptions and behavior. Indeed, it is possible that the timing of our two studies (i.e., 2024) may have influenced how participants perceived their own and others' use given the dynamism of the social media ecosystem.

A second limitation concerns the self-reported nature of the measure of social media frequency. Self-reported measures of social media use only moderately correlate with logged measures of social media use (Parry et al., 2021). Thus, self-reports may be inaccurate (or at least noisy) measures of how much people actually use social media. Self-

reported social media use may also correlate with perceptual outcome variables because such reports are capturing the construct of perceived use rather than actual use (Sewall et al., 2020). In the context of the present findings, the relationships between participants' self-reported use and their perceptions of others' use may have appeared stronger than they actually are. Future work may seek to observe whether the present findings replicate using measures of logged use.

A related direction is whether inaccuracies in the perception of others' media use are greater than inaccuracies in perceptions of one's own media use. To do so, future studies may compare logged and self-reported media use. Such relationships may be complicated by the platform. For example, prior work showed that people are less accurate at estimating their use on platforms used in a more fragmented, intermittent way (Verbeij et al., 2021). Comparisons of self-report versus logged social media use in further work will accordingly benefit from cross-platform analysis. In addition, because the present research focused on general use frequency, future research may also examine the relationship between specific platform-based behaviors and perceptions of others' use.

Finally, there are limitations in the generalizability of our samples. Our studies used college student samples in the United States, who do not inhabit the same social media ecologies as the rest of the population. Most of the participants in our studies were also women, and thus the findings may be biased toward their behavioral patterns (e.g., Tifferet, 2020). Future research may examine whether the findings replicate in other age or generational groups, as well as in other countries. Beyond generalizability, one direction for future research is exploring how different groups perceive each other's social media use. Our data suggest, for example, that college students have inaccurate views of other Americans' use, especially for Twitter. College students also showed the least egocentric projection of their social media use onto other Americans in terms of Twitter. An open question is whether older adults have corresponding (mis)perceptions of younger users in terms of certain platforms (e.g., TikTok). Such investigations will help untangle whether egocentric projection effects vary due to specific platform features or affordances, as opposed to or in addition to demographic use patterns.

Conclusion

Social media platforms offer people new viewpoints into their social worlds and the social worlds of others, but those views may also facilitate social misperceptions. Going beyond the aphorism "social media is not real life," we quantified the extent to which people displayed misperceptions of others' social media use (i.e., descriptive norms), while also considering the moderating role of egocentric biases. Our findings lay the groundwork for future researchers to investigate the interwoven factors of social norms, cognitive biases, and media behaviors. Most crucially, more frequent users of social platforms were more likely to overestimate others' use of those platforms, and these overestimations were also associated with perceiving that others see the same content. Our studies thus illustrate how social media can act as windows into others' experiences by connecting user frequency to perceptions of others' use and what they see. For some, social media may facilitate a reasonably accurate sense of what others' behaviors and experiences are like. For others, social media has the potential distort users' perceptions of other people. In sum, despite the expanding surface area of the social

media ecosystem, we find that people tend to look through their own social media windows and think others do too.

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